AXIOM: A Correctness-Aware Graph-Attention Architecture for Federated Multi-Agent Slice Admission Control

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Abstract

Fifth generation (5G) network slicing must admit heterogeneous service requests across many cooperating base stations under strict per-slice guarantees. This paper presents AXIOM, a correctness-aware graph-attention architecture for federated multi-agent slice admission control. A shared Graph Attention Network (GAT) encoder reads the base-station cluster graph and feeds decentralised per-slice action heads. The policy trains centrally and executes locally, so each gNodeB (gNB) decides independently from a shared view of the cluster. The design closes three gaps identified in our prior systematic review, namely trustworthy evaluation, privacy-preserving coordination, and validated deployment. On the metric that reflects the training objective, AXIOM decisively outperforms an independent-learner baseline. Scaled from 3 to 19 gNBs, it resists a training collapse that overwhelms a flattened single-agent baseline. It converges across 64 independent seeds to a stable 35.4% [25.5%, 45.8%] residual rate already present at 7 gNBs. Federated training adds no measurable reward cost, and differential privacy degrades decision quality only beyond a clear threshold. A correctness-aware metric pair and a reproduction-versus-replication discipline keep every reported result trustworthy. A live deployment on OpenAirInterface (OAI) hardware confirms that the offline-trained policy executes correctly end to end, establishing deployability on a real Open Radio Access Network stack. Contention-dependent behaviour is characterised in a disclosed offline stress regime calibrated harder than the live rig reaches.

Keywords: Network Slicing, Slice Admission Control, Multi-Agent Reinforcement Learning, Graph Attention Networks, Federated Learning, Open RAN

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1 Introduction

Network slicing virtualises physical radio access network (RAN) to serve heterogenous service delivery under Ultra-Reliable Low-Latency Communication (URLLC), enhanced Mobile Broadband (eMBB), and massive Machine-Type Communication (mMTC) categories. Each slice carries distinct Service Level Agreement (SLA) targets that are in turn based on Quality of Service (QoS) thresholds. Our earlier work [7, 8] validated that a Deep Q-Network (DQN) admission controller can enforce SLAs for a single gNodeB (gNB) in a live OpenAirInterface (OAI) testbed. However, a real deployment spans multiple gNBs that share backhaul, spectrum planning, and a cluster-wide load-balance objective. Thus, slice admission at one gNB is rarely isolated, and coordinated admission is in fact a multi-gNB, multi-agent challenge. This research aims to address whether a shared, graph-structured view of the cluster could enable per-gNB agents to decide better than a single agent on a flattened state, as opposed to independent agents. Fig. 1 places the research premise in context.

This paper introduces AXIOM, an Attention-based eXecution with Integrity for O-RAN Multi-agent slice-admission controller. AXIOM is a centralised-training, decentralised-execution multi-agent architecture that is built on a shared Graph Attention Network (GAT) [17] encoder over a multi-gNB topology, which feeds decentralised per-slice Q-heads. We deliver it together with the evaluation discipline needed to trust what it reports. Three gaps from our systematic review of 67 core studies motivate this pairing, as Section 2 details. First, no reviewed study checks whether its own outcome metric can score a degenerate policy as well as a correct one. Second, none replicates a headline architectural finding on an independent seed sample. Third, none tests whether a live testbed can reach the contention regime its offline claims assume. We close all three gaps through a per-component realisation of the architecture, namely the GAT encoder, the Multi-Agent Reinforcement Learning (MARL) controller, the Federated Learning (FL) coordinator, and the Differential Privacy (DP) layer, each measured and isolated rather than folded into one end-to-end number. We apply a correctness-aware metric pair and a reproduction-versus-replication discipline throughout, and we add a live single-gNB deployment that tests, rather than assumes, whether the rig reaches genuine per-slice scarcity.

We extend the architecture to federated training with differential privacy, and we stress it under mid-episode gNB dropout, demand spikes, agent churn, and cluster growth to $N \in \{7, 19\}$ gNBs under ring and hex-grid adjacency. On the objective it trains on, AXIOM decisively beats an independent-learner ablation and resists a training collapse that becomes total for a flattened single-agent baseline. A correctness-aware metric pair and an independent-seed replication keep every reported result trustworthy.

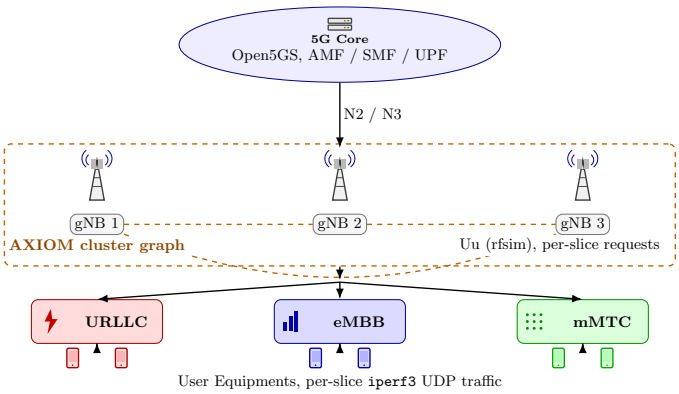


Fig. 1 System view. A cluster of gNBs, coordinated by the shared AXIOM encoder over the cluster graph and backhauled to a 5G core, serves UEs across the URLLC, eMBB, and mMTC slices. Each gNB admits or rejects per-slice requests through a PRB-ratio ceiling.

2 Related Work

Our prior systematic review [9] screened 6,286 records down to 67 core studies from 2024 to 2026 on machine-learning-driven, user-centric distributed network slicing. The review organised them around three enabling pillars, namely topology-aware graph representation, distributed multi-agent control, and privacy-preserving federation. Each pillar is well developed on its own, but the three are rarely combined under realistic constraints, and never with a validated live deployment. No reviewed study checks whether its own evaluation metric can mask a degenerate policy, and none replicates a headline finding on an independent seed sample, which is the gap this paper fills.

Several recent works come close on a single axis of this problem. Shao et al. [14] motivate our GAT encoder choice by showing that graph attention strengthens inter-agent cooperation over plain DQN and Advantage Actor-Critic (A2C) baselines. Ahmadi et al. [2] study topology generalisation through graph attention, which complements but does not answer the fixed-topology evaluation-integrity question we ask in Section 5.4. Sulaiman et al. [15] are the closest prior art to our joint slicing-and-admission-control problem. None of these three checks whether its reward-optimal policy could be mistaken for a degenerate one, and none replicates a headline result on an independent seed sample.

Robustness studies share our concern but stop short of it. Tashman and Cherkaoui [16] and Fatehi et al. [5] study robustness to a disruptive condition, as our disruption campaign in Section 5.3 does, yet neither checks whether the reported robustness metric is trustworthy and neither replicates a finding on an independent sample. That is the gap our independent-seed replication closes, and the need is concrete, since replication overturned our initial churn-immunity reading. Zhang et al. [19] are the closest periodic-synchronisation precedent, and Yasin et al. [18] are the closest precedent for federated learning with differential privacy. Their threat model differs from ours, since they secure the controller rather than characterise a control policy's own privacy-utility trade-off, which our σ sweep in Section 5.2 reports, and neither

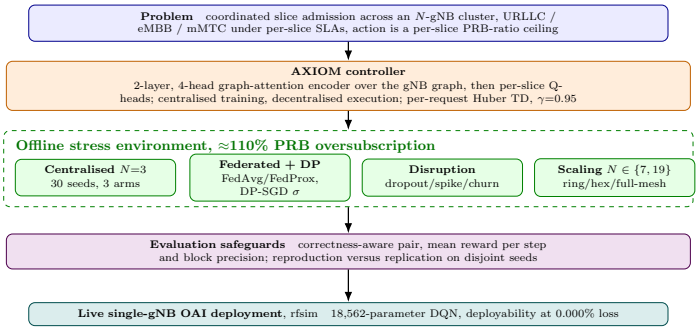


Fig. 2 Overview of the research. An AXIOM controller is evaluated across four offline campaigns in a stress environment, every result checked by two safeguards, then validated on a live single-gNB testbed.

adds a correctness-aware check on whether federating or privatising changed decision quality. A further, closely related set [3, 6, 12, 13] combines two of the three pillars, yet like every work above, each evaluates on its own reported outcome metric, computed once, on one seed or run.

Rather than adding a new point in the GAT, MARL, and FL design space, this paper fills that gap with a demonstration on AXIOM, extended to federation, privacy, and disruption. It shows that two checks are both necessary, namely testing a metric's own trustworthiness and replicating a finding on an independent sample. Skipping either changes what a reader may believe the resulting numbers mean.

3 System Model and Problem Formulation

Fig. 2 provides an overview of the system. The AXIOM controller runs across four offline campaigns in a disclosed stress environment. Two correctness safeguards check every result. The offline-trained logic is then verified on the live single-gNB testbed of Fig. 3.

We consider a cluster of N=3 gNBs, each serving the same three slices, URLLC, eMBB, and mMTC, under a shared per-gNB Physical Resource Block (PRB) budget B. An admission decision is a PRB-ceiling adjustment through the `slicing_control_m` minimum and maximum ratio, rather than a literal per-flow accept or reject, since the real OAI xApp control interface exposes only ratio ceilings. No real multi-gNB physical topology is available to us, so we take the least assumption-laden choice and treat the cluster as fully connected, as a modelling choice rather than a measured fact. At each control step, every gNB holds zero or more pending admission requests, each tagged with a slice identity, and an agent decides whether to accept or reject for each request independently, conditioned on the current cluster state. The discrete action space matches our single-gNB work exactly at `action_dim`=2, so any coordination benefit found here is due to the architecture and not a richer action space.

The step reward follows our prior single-gNB formulation. It has a per-slice service term net of SLA-violation penalties, a congestion charge on admitted volume, and a load-imbalance penalty across gNBs. The step reward r_t sums over slices indexed by k. It uses priority weight ω_k, per-accept reward R_k, admitted count n_k, violation

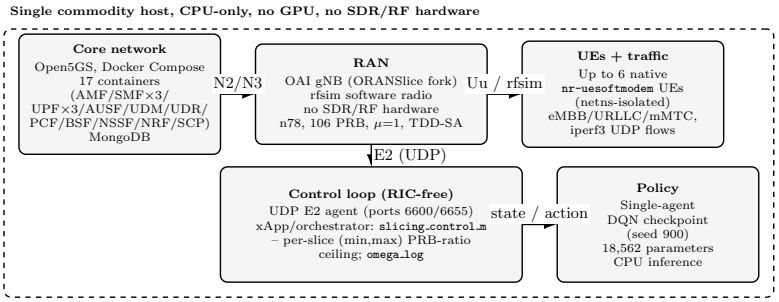


Fig. 3 Real testbed stack, a single-host, software-radio rfsim, RIC-free O-RAN deployment distinguished from SDR hardware-in-the-loop testbeds. Every block and label is a measured or configured value from this rig.

penalty λ_k, and the indicator $\mathbf{1}[\cdot]$ equal to one exactly when slice k is in SLA violation that step,

$$r_t = \sum_k \big(\omega_k R_k\, n_k - \lambda_k\, \mathbf{1}[v_k]\big) \;-\; \kappa\, C_t \|a_t\|_1 \;-\; \beta\,(1 - F_t), \tag{1}$$

where $\|a_t\|_1$ is the total admitted this step, C_t the cluster-mean congestion level, κ the congestion coefficient, $F_t \in [0, 1]$ the min-over-max PRB-utilisation fairness ratio, and β the load-balance weight. The violation flag v_k follows from a continuous SLA margin m_k. The correctness-aware metrics of Section 4.4 also build on this margin. We take it as the worse of the queue and loss channels,

$$m_k = \min\Big(1 - \tilde{q}_k,\; 1 - \tfrac{\ell_k}{\ell_k^{\max}}\Big), \qquad v_k \Leftrightarrow m_k \le 0, \tag{2}$$

with $\tilde{q}_k$ the slice backlog relative to its budget, and ℓ_k and $\ell_k^{\max}$ its loss proxy and loss budget. These weights are tuned in prior work so that, under congestion, the reward-optimal policy is differentiated shedding, which continues accepting URLLC, mostly accepts eMBB, and rejects mMTC. We confirm this by Q-value inspection on a working baseline policy in Section 4.1, rather than assume it. Every result uses an offline, closed-loop simulation, except the bounded live anchor of Section 5.5. The simulation is deliberately oversubscribed to roughly 110% of one gNB's PRB budget and reuses our existing multi-gNB configuration unmodified, as a deliberately harder, disclosed stress environment. The concrete rig behind every live measurement is the single-host, software-radio, Open Radio Access Network (O-RAN) stack of Fig. 3, which runs without a near-real-time RAN Intelligent Controller (RIC).

4 The AXIOM Architecture

4.1 Shared Encoder, Decentralised Q-Heads

Fig. 4a shows the architecture. At every control step, each gNB forms its own per-slice feature row x_i. The features are PRB-used-ratio, congestion-level, and queue-length-norm. These rows stack into a joint node-feature matrix $X \in \mathbb{R}^{N \times d}$ over the N

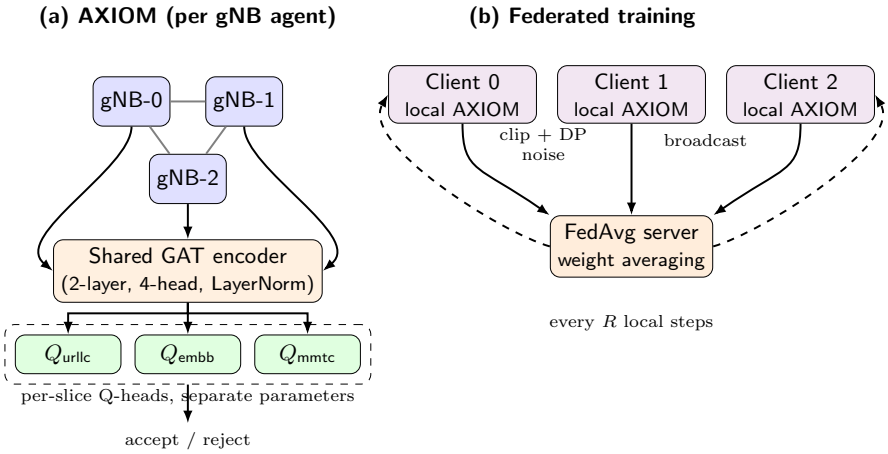


Fig. 4 AXIOM architecture. a, A shared graph-attention encoder over the fully-connected 3-gNB topology produces one embedding per node, consumed at execution time by per-slice Q-heads using only that node's row. b, The federated variant, each gNB a client synchronised by periodic FedAvg with an optional DP-SGD clip-and-noise step.

gNB nodes, with d features each. A two-layer, H=4-head GAT [17] produces one embedding per gNB. Each head h decomposes additive attention rather than scoring a concatenation. It uses a learned source vector a_s^h and destination vector a_d^h, applied to the shared linear projection W^h. Here ϕ is the leaky rectifier of slope 0.2, and $\mathcal{N}_i$ is the set of neighbours of node i under the adjacency mask,

$$\varepsilon_{ij}^h = \phi\big(a_s^{h\top} W^h x_i + a_d^{h\top} W^h x_j\big), \qquad \alpha_{ij}^h = \frac{\exp \varepsilon_{ij}^h}{\sum_{l\in\mathcal{N}_i} \exp \varepsilon_{il}^h}. \tag{3}$$

Here ε_{ij}^h is the unnormalised attention score from node i to neighbour j in head h, and α_{ij}^h is its normalisation over i's neighbours, with l ranging over $\mathcal{N}_i$. Hidden layers concatenate the H heads under an ELU nonlinearity, while the final layer averages them. A per-layer normalisation operator $\mathrm{LN}(\cdot)$ then rescales each node's embedding to zero mean and unit variance, and produces the final per-gNB embedding z_i,

$$z_i = \mathrm{LN}\Big(\tfrac{1}{H}\textstyle\sum_{h=1}^{H}\sum_{j\in\mathcal{N}_i} \alpha_{ij}^h\, W^h x_j\Big). \tag{4}$$

Centralisation comes from this encoder, whose parameters every agent's loss updates jointly, so it sees the full cluster state during training. Decentralised execution comes next, since at action-selection time each agent consumes only its own embedding row z_i to decide its own pending requests.

The `LayerNorm` and the per-slice heads both answer a real training collapse, rather than a theoretical preference. An early design fed each embedding, concatenated with a one-hot slice context, into a single shared Q-head multi-layer perceptron (MLP). All three slices then update the same shared parameters. URLLC's large priority weight

Algorithm 1 AXIOM: centralised training, decentralised execution

Require: encoder f_θ, per-slice Q-heads $\{Q_{\phi_k}\}$, targets $f_{\theta^-}, \{Q_{\phi_k^-}\}$, buffer $\mathcal{D}$, adjacency A, discount γ, sync period C

for each step t, observing joint node features X_t **do**
  $E_t \leftarrow f_\theta(X_t, A)$ {centralised, one row per gNB}
  for each pending request at gNB i, slice k, pick $a \leftarrow \arg\max_a Q_{\phi_k}(E_t[i])[a]$, ϵ-greedy {decentralised, own row only}
  step env, store $(X_t, X_{t+1}, \{(i,k,a)\}, r_t, \text{done})$ in $\mathcal{D}$
  if $|\mathcal{D}| \geq$ warmup **then**
    sample minibatch B; $E_B \leftarrow f_\theta(X_B, A)$, $E'_B \leftarrow f_{\theta^-}(X'_B, A)$
    per request, $y \leftarrow r + \gamma(1-\text{done})\max_{a'} Q_{\phi_k^-}(E'_B[i])[a']$
    $\mathcal{L} \leftarrow$ mean per-request Huber loss, never summed per step; update $\theta, \{\phi_k\}$, clip grad-norm 1.0
  end if
  every C episodes, $\theta^- \leftarrow \theta$, $\phi_k^- \leftarrow \phi_k$
end for

produces large gradients that overwrite mMTC's much smaller true signal. This is a standard shared-parameter class imbalance, and it drove the evaluation policy to block zero requests. We replace the single shared head with one small MLP per slice, selected by the context one-hot. Each slice then owns its parameters, so one slice's gradients can no longer overwrite another's. Together with the `LayerNorm`, this reduced the collapse from 30 of 30 to 9 of 30 seeds. Correctly-targeted differentiated shedding rose from 0 to 21 of the 30 seeds. A seed here is one independent random initialisation of network weights and environment stream. We report every seed rather than a single best run.

Algorithm 1 states the training loop, where the encoder updates from every request in a sampled minibatch regardless of gNB or slice. This is centralised training, while action selection at rollout time consults only the acting agent's own embedding row, which is decentralised execution. Every pending request at gNB i, slice k is its own temporal-difference sample, and we never sum requests across a step before the loss. For a request with immediate reward r, next-state embedding z'_i, and candidate next action a', the bootstrapped target y at discount γ and the minibatch loss are

$$y = r + \gamma\,(1-d)\max_{a'} Q_{\phi_k^-}\big(z'_i\big)[a'], \qquad \mathcal{L} = \frac{1}{|B|}\sum_{(i,k,a)\in B} \ell_\delta\Big(Q_{\phi_k}(z_i)[a] - y\Big), \tag{5}$$

where Q_{ϕ_k} is slice k's online Q-head, $Q_{\phi_k^-}$ its target copy, d the episode-done flag, and B the sampled minibatch of requests. We use the Huber loss ℓ_δ in place of squared error, which keeps the deeper encoder stable under long bootstraps.

4.2 Federated Training with Differential Privacy

Fig. 4b shows the federated variant. Each gNB becomes a federated-learning client. It holds a private copy of the same network and trains only on its own requests. Transitions are partitioned by owning agent, so no client sees another's raw admission

data. Each client's GAT encoder still consumes the full joint node-feature snapshot. Every R local steps, the server averages all clients' online weights by FedAvg [11] and broadcasts the result back. We also implement FedProx [10], which adds a proximal term that pulls local weights toward the last broadcast global model. We sweep it at $\mu \in \{0.01, 0.1, 1.0\}$ under both homogeneous and heterogeneous per-gNB load. FedProx earns no measurable dividend at any strength, since per-round client drift is negligible relative to the loss scale.

Differential privacy gives a formal guarantee that one client's training data cannot be inferred from the shared model. We apply it as a standard differentially private stochastic gradient descent (DP-SGD) step [1]. Each client clips its gradient to a fixed norm C_{dp}. When privacy is enabled, the client then perturbs the gradient with Gaussian noise scaled by a noise multiplier σ, before its optimiser step. We report σ as the primary privacy parameter. Alongside it we report a formal (ε, δ) bound via zero-concentrated differential privacy (zCDP) [4]. Here ε and δ are the standard privacy-loss and failure-probability parameters. Composing the Gaussian mechanism over m per-client gradient releases gives the zCDP parameter ρ and the bound

$$\rho = \frac{m}{2\sigma^2}, \qquad \varepsilon(\delta) = \rho + 2\sqrt{\rho \ln(1/\delta)}. \tag{6}$$

We apply the same clip whether or not noise is added, so a σ=0 run and a σ>0 run differ only in whether noise is injected, which is what lets the privacy-cost comparison in Section 5.2 isolate the effect of the noise. Each federated round therefore runs R local Huber-TD steps per client with an optional FedProx proximal term $\frac{\mu}{2}\|\theta_i - \bar{\theta}\|^2$, clips and optionally perturbs the gradient, and closes with a FedAvg average that is broadcast back.

4.3 Experimental Setup

Both campaigns use 300 training and 50 evaluation episodes per seed. The centralised campaign uses a fixed 30-seed list, seeds 900 to 929, across three arms. The first arm is AXIOM, which the result figures label GAT-CTDE after its architecture. The second is a single-agent DQN baseline run against the flattened joint state. The third is an independent-DQN ablation. An ablation is a deliberately reduced variant with one component removed, so its effect can be isolated. Here it is one per-gNB DQN with no parameter sharing and no topology information, which isolates the contribution of the shared encoder and centralised training. All three arms use matched hyperparameters, with discount factor $\gamma = 0.95$ and per-episode epsilon decay. The federated campaign uses the first 10 of these seeds and a five-level noise sweep $\sigma \in \{0, 0.5, 1, 2, 4\}$. Its σ=0 level is therefore directly comparable to the same 10 centralised seeds. All confidence intervals (CIs) are 95% bootstrap percentile intervals over 10,000 resamples, and paired comparisons use the Wilcoxon signed-rank test. The remaining settings are fixed across every seed. AXIOM trains at learning rate $1{\times}10^{-4}$ and the DQN baselines at $1{\times}10^{-3}$, both under the Adam optimiser, with epsilon decaying from 1.0 to 0.05 at a per-episode factor of 0.985 and the target network synchronised every 10 episodes. The GAT encoder uses hidden and output dimension 16, feeding per-slice Q-heads of hidden

Table 1 Four evaluation-pipeline failures this paper's safeguards caught, none of which a standard pipeline would have surfaced.

Failure	Missed by	Caught by	Outcome
Training collapse (§4.1)	Compliance blind to no-op	Correctness-aware pair	0/30→21/30 differentiate
Eval-log contamination	Append-mode log stacking	Episode-index audit	Logs cleared, no retrain
Reproducibility gap (§5.3)	Single-sample overclaim	Repro-vs-replication	Churn claim retracted
Scheduler-floor bug (§5.5)	Can't see cross-layer floor	Direct state instrumentation	>100k readings clear

dimension 32. Each federated round runs 50 local steps under FedAvg with FedProx weight μ=0 and a differential-privacy gradient-clip norm of 1.0.

4.4 Evaluation Safeguards

Two safeguards make the results below trustworthy. Table 1 summarises the four pipeline issues they caught. A standard pipeline would have surfaced each one as a clean result. The first safeguard is a correctness-aware metric pair, computed from data every run already logs. Mean reward per step is the actual training objective. Block precision is the fraction of blocked requests that correctly target mMTC, the only slice the reward calibration ever makes reject-optimal. Writing b_{mMTC} and b for the mMTC-targeted and total blocks over an evaluation,

$$\pi = \frac{b_{\mathrm{mMTC}}}{b}, \quad b > 0, \qquad \text{undefined for } b = 0, \tag{7}$$

a seed that never blocks is therefore excluded, rather than scored as a success.

The pair is needed because `sla_compliance_all_slices` scores worse when the policy becomes more correct. Blocking mMTC under congestion never recovers mMTC's own SLA margin in this stress regime. We re-checked every paired comparison in this paper against both metrics. The correctness-aware pair and compliance disagree on direction or significance in 44 of 49 comparisons, or 89.8%. A disjoint supplementary seed range confirmed this at 21 of 24. Every number in this paper that matters is therefore reported through the correctness-aware pair.

The second safeguard is a reproduction-versus-replication discipline, in which we recompute every headline comparison on an independent, disjoint seed sample and report both. Reproduction re-runs the same seeds, whereas replication draws fresh ones, so only replication can catch a result that is an artifact of one seed sample. The remaining pipeline issues of Table 1 were each fixed at their source, with the churn correction detailed in Section 5.3 and the scheduler floor in Section 5.5.

5 Results

We compare three policies throughout this section. AXIOM is the architecture of this paper, a shared graph-attention encoder that reads the whole base-station cluster and

Table 2 How the three policies compare. AXIOM is the architecture of this paper; the two DQN policies are ablations that strip out its coordination. The three-gNB values come from the campaign of Table 3.

Property	AXIOM	Single-agent DQN	Independent DQN
Design	Shared graph encoder, one head per slice	One flattened controller, no agent structure	One learner per gNB, no sharing
Mean reward per step, 3 gNBs	Highest, 14.062	Close, 13.867	Lowest, 12.601
Block precision, 3 gNBs	High, 0.952	Highest, 0.987	Low, 0.901
Behaviour at 19 gNBs	Partial collapse, 35.4%	Total collapse, all 15 cells	Collapses rarely, 0.0–16.7%
Reads cluster layout	Yes	No	No
Federated and privacy-ready	Yes, no measurable reward cost	Not applicable	Not applicable

feeds one small decision head per slice. The single-agent DQN flattens the cluster into one controller that sees no per-gNB structure. The independent DQN gives each gNB its own learner that never looks at its neighbours. The two DQN policies are deliberate ablations, so any gap between them and AXIOM isolates what the shared encoder and joint training actually add.

Three quantities carry the comparison. Mean reward per step tracks the objective the policies train on, so higher is better. Block precision is the share of a policy's rejections that land on the slice the reward rule marks lowest priority, so a value near one means the policy sheds the right traffic. Collapse names a failure mode in which a trained policy stops discriminating and simply accepts every request, which can look acceptable on raw reward yet abandons admission control altogether. Table 2 sets out how the three policies stand on these axes, and the subsections that follow give the evidence.

The headline is that no baseline matches AXIOM on every axis at once. The single-agent DQN rivals it at three gNBs and even edges its block precision there, yet collapses completely at nineteen. The independent DQN avoids total collapse but learns the weakest policy of the three. AXIOM is the only policy that earns the top reward, blocks accurately, and still functions as the cluster grows, which is the sense in which it stands above the other two.

5.1 Centralised

The three-gNB fully-connected setting is deliberately the hardest case for a graph encoder, since every gNB already sees every other and there is little topology left to exploit. It is therefore a clean place to establish the evaluation-integrity findings with no topology effect in play. The encoder's coordination advantage is tested in earnest only once the cluster grows and its adjacency thins, in Section 5.4.

Table 3 and Fig. 5 report the final thirty-seed campaign. AXIOM earns the highest mean reward per step of the three policies. Its lead over the independent-DQN ablation is large and statistically clear, at +1.461 with 95% CI [0.550, 2.782] and $p < 0.0001$,

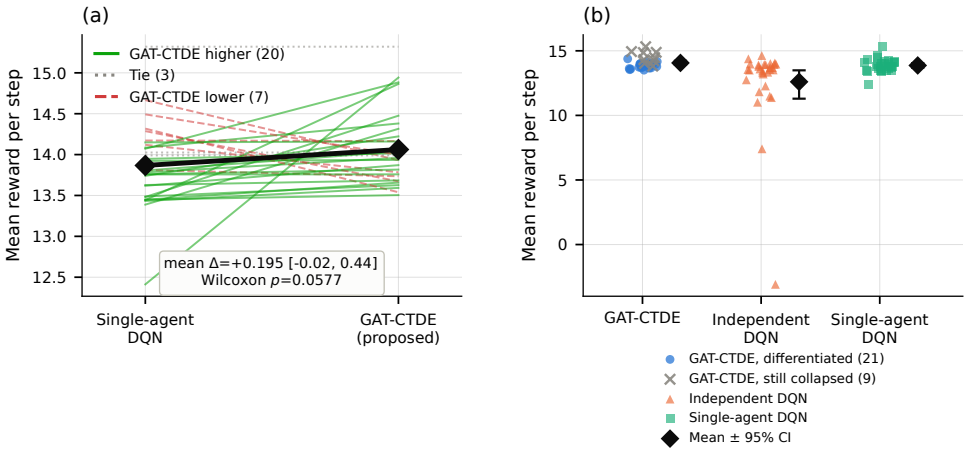


Fig. 5 M2 centralised campaign, per-seed evidence. a, Paired per-seed mean reward per step versus single-agent DQN, coloured by win, tie, or loss, the source of the borderline $p = 0.0577$. b, Per-seed reward distribution for all three arms.

Table 3 M2 campaign results, 30 seeds/arm, 95% bootstrap CIs.

Arm	Mean reward/step	Block precision	Compliance*
AXIOM	14.062 [13.91, 14.23]	0.952	0.151 [0.083, 0.225]
Indep. DQN	12.601 [11.30, 13.49]	0.901	0.049 [0.013, 0.093]
Single-agent	13.867 [13.69, 14.05]	0.987	0.115 [0.061, 0.176]

* Low compliance is the metric artifact Section 4.4 explains, not a performance failure.

which pins a real gain on the shared encoder and joint training. Its lead over the single-agent DQN is genuine but small, at +0.195 with 95% CI [−0.017, 0.442], and at thirty seeds it does not reach significance at $p = 0.0577$. We state that limit plainly rather than overclaim it. On block precision AXIOM blocks the reward-correct slice with precision 0.952, just behind the single-agent policy at 0.987 and well ahead of the independent learner at 0.901. The compliance metric points nowhere, at +0.036 with $p = 0.4549$, and that is the measurement artifact Section 4.4 explains rather than any real performance gap.

5.2 Federated Training and DP

With federation switched on and no privacy noise, mean reward per step barely moves against the centralised run on the same ten seeds, by +0.133 with 95% CI [−0.138, 0.470] and $p = 0.8125$. Federating the training therefore costs nothing measurable, and both variants keep perfect block precision on the seeds that block at all. We then add differential-privacy noise, random perturbation to each shared update that hides any single gNB's data. Block precision holds perfect through a noise multiplier of σ=1.0,

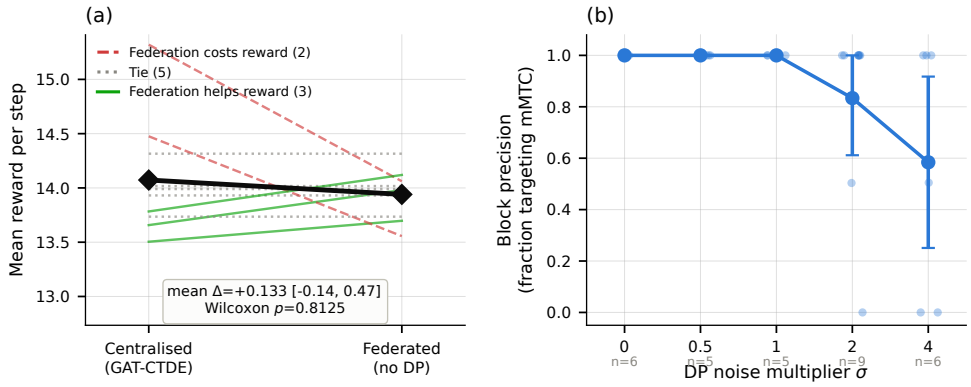


Fig. 6 M3 federated and DP sweep. a, Paired per-seed federation cost on mean reward per step, no measurable difference at $p = 0.8125$. b, Block precision versus DP noise multiplier σ, perfect through σ=1.0 then dropping sharply.

then falls sharply to 0.834 at σ=2.0 and 0.584 at σ=4.0, as Fig. 6b shows. This is a clear threshold rather than a gentle slide. Mean reward per step stays inside a narrow 12.45–14.19 band throughout, because the noise erodes which slice gets blocked long before it erodes whether the policy decides at all. Read together with the FedProx null result, this answers a deployability question directly. Periodic, infrequent synchronisation is enough for this architecture, a disaggregated controller needs no per-step coordination between gNBs to match centralised reward, and privacy can be bought at a bounded and known precision cost.

5.3 Disruption

We test every policy's frozen checkpoint against three mid-episode disruptions, injected at evaluation time with no retraining. A 'gNB dropout' forces one gNB to reject everything and blanks its row in the shared feature matrix for a window. A 'demand spike' briefly multiplies the arrival rate from outside the policy. An 'agent churn' swaps one gNB's decisions for a fresh, untrained copy of its own policy for a window and leaves the others intact. Churn applies to every genuinely multi-agent policy but not to the single-agent one. For dropout and churn the severity is how long the window lasts, at 10%, 30%, or 60% of the episode. For the spike it is the arrival multiplier, at 2×, 4×, or 8×. Because raw reward grows mechanically with request volume, we report a volume-normalised reward $\bar{r}$ over an episode of T steps, dividing each step reward r_t by its own pending-request count n_t before averaging,

$$\bar{r} \;=\; \frac{1}{T}\sum_{t=1}^{T}\frac{r_t}{\max(n_t,\,1)}. \tag{8}$$

Across both dropout and churn the cost accelerates with severity rather than rising in proportion, with 9 or 10 of every 10 seeds hurt and Wilcoxon $p \leq 0.03$. For AXIOM

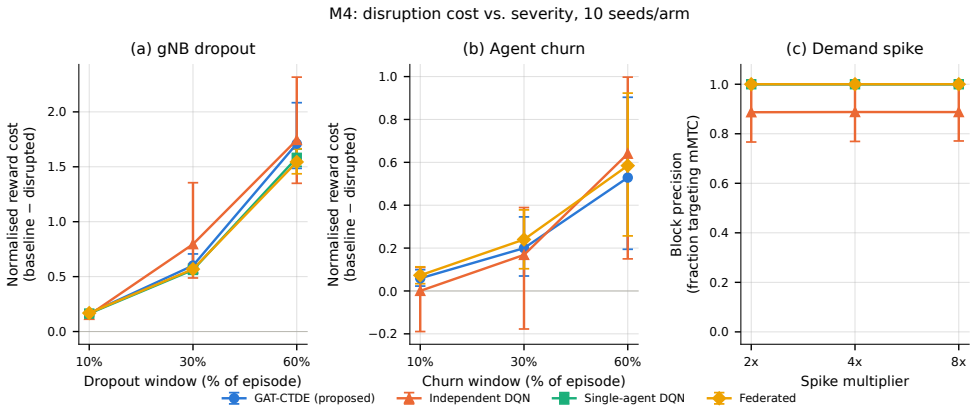


Fig. 7 M4 disruption campaign, 10 seeds per arm. a, b, Normalised reward cost versus severity for dropout and churn, accelerating with severity. c, Block precision versus demand-spike multiplier, flat and near-ceiling.

the dropout cost grows 0.165, 0.601, 1.710 across the three severities, and the churn cost grows 0.059, 0.200, 0.529. Every policy shows the same accelerating shape, so this is a property of the task rather than of any one design. The demand spike leaves decision quality intact on the volume-invariant block-precision metric in Fig. 7c, so its apparent reward gain is a counting effect, not a better policy.

One disruption result needed a second, independent sample to report honestly. An early pass read the independent DQN as immune to churn, since it never attends to another agent's features, but that rested on a small and non-significant ten-seed sample. A disjoint replication overturns it, with a churn cost that is significant at every severity and in fact larger than the coordinated policies', rising 0.097, 0.657, 1.223 against the earlier 0.000, 0.168, 0.640. The honest conclusion is that churn hurts every policy, and that the earlier immunity claim was a small-sample artifact. This was the single correction among four headline cross-section comparisons re-run against independent samples. The other three held with overlapping intervals, namely AXIOM over the independent DQN on reward at $+1.461$ then $+0.767$ with $p < 0.0001$ in both, AXIOM over the single-agent policy at $+0.195$ then $+0.445$, and a negligible federation cost in both.

5.4 Cluster-Size Scaling

We grow the cluster from three gNBs to seven and nineteen under three layouts, fully-connected, ring, and hex-grid, scaling the synthetic arrivals with the cluster so per-gNB load stays roughly constant. Only AXIOM's decision head actually reads this layout, so the two DQN policies double as a check that nothing depends on topology by accident. We normalise reward per gNB as in Eq. 8, and we lean on block precision rather than compliance here, because compliance is combined across gNBs with a logical OR and drifts to zero as the cluster grows whatever the policy does. The main

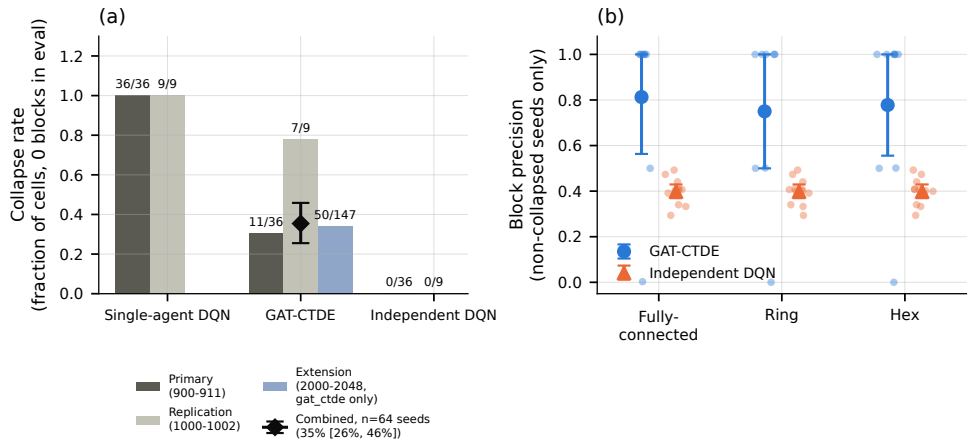


Fig. 8 N=19 cluster, per-seed evidence. a, Collapse rate pooled across topologies, single-agent DQN total and AXIOM converging to 35.4% across 64 seeds. b, Block precision by topology for non-collapsed seeds.

sample is twelve seeds at nineteen gNBs across every layout and policy, with further independent seeds to test generalisation.

At nineteen gNBs the per-gNB reward margins are not significant for any layout, and that is expected rather than disappointing. The single-agent policy's always-accept collapse banks the service reward on every request, while AXIOM pays a real price each time it correctly rejects low-priority mMTC traffic. The true separation therefore shows up in collapse resistance and block precision, not in raw reward.

The contrast between the three policies at scale is stark. The single-agent DQN collapses completely at nineteen gNBs, in all 15 of 15 cells across both samples and every layout, which a direct probe of its decision values confirms at the level of individual choices. AXIOM, by contrast, collapses only part of the time. The independent DQN never collapses fully, but settles into a narrow and mediocre 0.29–0.49 block-precision band that is identical across layouts, exactly as expected for a policy that never reads its neighbours.

Pooled across 64 independently drawn seeds, covering 68 of 192 cells, AXIOM's collapse rate converges to a stable value,

$$\hat{p}_{\text{collapse}} = 35.4\%, \qquad 95\%\ \text{CI}\ [25.5\%,\ 45.8\%], \tag{9}$$

This single bootstrapped estimate replaces two smaller samples that had disagreed. A properly powered twelve-seed run at seven gNBs, with a disjoint twelve-seed replication, puts AXIOM at 41.7% and 58.3% collapse across the two samples. Both are statistically indistinguishable from Eq. 9, so this collapse-proneness is present from seven gNBs onward, not only at large scale. Over the same two samples the single-agent DQN collapses at 91.7% and 75.0%, against 0% in its earlier three-seed pilot, the sharpest single result in this extension, while the independent DQN collapses at

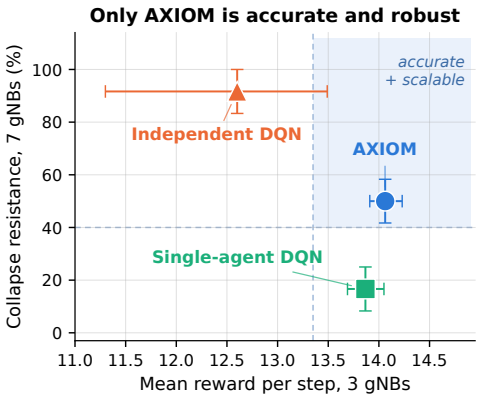


Fig. 9 Reward quality against scaling robustness, from mean reward per step at three gNBs with 95% CIs and collapse resistance at seven gNBs across two 12-seed samples. AXIOM alone is both accurate and robust, while the other two each miss one axis.

0.0% and 16.7%, confirming that it collapses rarely rather than never. AXIOM therefore sits between the two, far more robust than the single-agent policy it is meant to replace, and the only one of the three that stays robust while also learning a strong, topology-aware blocking rule. Fig. 9 places the three policies on one map of reward quality against scaling robustness. AXIOM alone falls in the region that is both accurate and robust, while the independent DQN trades reward away for its robustness and the single-agent policy trades robustness away for reward.

5.5 Live Deployment

This section reports what the real OAI testbed of Fig. 3, introduced in Section 3, can and cannot establish about the offline-trained policy. We run a single-agent-DQN checkpoint, seed 900, against a real gNB, real User Equipments (UEs), and real per-slice User Datagram Protocol (UDP) traffic, at one UE per slice and again at two UEs per slice.

A capacity probe at a deliberately wide-open ceiling showed all three slices serving essentially 100% of offered demand with 0.000% loss at both loads. The frozen checkpoint's decision logic runs correctly end to end, through the real E2 control loop, with real accept and reject decisions, and its signature low-priority-only blocking of mMTC reappears at both loads, which establishes deployability end to end. Because the rig never actually runs short of capacity, the same pattern is equally consistent with the policy's trained priority bias, so we claim admission-control behaviour under contention only in the disclosed offline stress regime, not live. Two integrity issues surfaced and were fixed during these checks. A scheduler-configuration error had turned the per-slice PRB-ratio ranges into raw grants below the gNB scheduler's hard five-PRB floor, which we fixed at the root, after which direct instrumentation confirmed zero below-floor grants across more than 100,000 readings. A separate unit mismatch in the live SLA-margin diagnostic was corrected by converting backlog into a queueing-delay estimate, using Little's law, against each slice's own latency budget.

6 Conclusion

This paper presented AXIOM, a correctness-aware graph-attention architecture for federated multi-agent slice admission control. A shared GAT encoder reads the cluster graph and feeds decentralised per-slice action heads. The design closes three gaps from our prior systematic review, namely trustworthy evaluation, privacy-preserving coordination, and validated deployment.

The architecture delivers on each of these fronts. AXIOM beats the independent-learner ablation decisively on the training objective, resists at 19 gNBs a collapse that overwhelms the single-agent baseline, and holds its reward under federation while differential privacy stays accurate up to a clear noise threshold. A live single-gNB deployment confirms that the offline-trained logic executes correctly on real hardware. These results carry honest bounds, since the multi-gNB, federated, and disruption campaigns are offline, the graph models only the gNB cluster, and the federated setting assumes a single operator.

Two safeguards keep these results trustworthy. A correctness-aware metric pair ties evaluation to the training objective, while a reproduction-versus-replication discipline tests each headline finding on an independent seed sample. Together they caught issues a standard pipeline would have reported as clean. The broader lesson is that evaluation here needs metrics tied to the objective, that architectural claims need an independent sample before they are trusted, and that a live testbed's reach must be checked rather than assumed.

Future work targets the remaining gaps from our review. The first goal is a full-scale simulation with real contention, so admission-control behaviour under genuine scarcity can be measured directly rather than in the disclosed stress regime. The second is Multi-Access Edge Computing (MEC) scaling, which adds compute-placement nodes to the cluster graph and extends the controller to joint admission and offloading. We also plan a live multi-gNB deployment, broader privacy settings across operators, and direct Quality of Experience (QoE) inference in the reward. Together these steps define AXIOM-X, the eXtended version of AXIOM, moving the architecture toward a scalable, privacy-aware, and user-centric controller for 5G and Beyond-5G networks.

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Declarations

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- **Competing interests:** The authors declare no competing interests.
- **Ethics approval and consent to participate:** Not applicable, since this study involves no human participants, animal subjects, or identifiable patient data. All

reported results are either offline simulation campaigns or measurements from a laboratory radio testbed.

- **Consent for publication:** Not applicable, since this manuscript contains no personal data or identifying information about any individual.
- **Data availability:** The experiment logs, checkpoints, and campaign result files underlying this paper's findings are available from the corresponding author on reasonable request.
- **Materials availability:** Not applicable, since this study involves no physical or biological materials.
- **Code availability:** The framework and experiment scripts used to produce this paper's results, including the reproducibility manifest and analysis scripts used to independently re-derive every table and figure in this paper, are available from the corresponding author on reasonable request.
- **Author contribution:** M. P. Kunasegran: conceptualisation, methodology, software, investigation, writing (original draft). W. L. Pang: supervision, writing (review and editing). S. K. Phang: supervision, writing (review and editing).

References

[1] Abadi M, Chu A, Goodfellow I, et al (2016) Deep learning with differential privacy. In: Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS), pp 308–318, https://doi.org/10.1145/2976749.2978318

[2] Ahmadi M, Moayyedi A, Sulaiman M, et al (2024) Generalizable 5G RAN/MEC slicing and admission control for reliable network operation. IEEE Trans Netw Service Manag https://doi.org/10.1109/TNSM.2024.3437217

[3] Bouroudi A, Outtagarts A, Hadjadj-Aoul Y (2025) GNN-based multi-agent DRL for energy-efficient multi-domain 6G resource orchestration. In: Proc. IEEE Global Commun. Conf. (GLOBECOM), pp 4457–4462, https://doi.org/10.1109/GLOBECOM59602.2025.11432814

[4] Bun M, Steinke T (2016) Concentrated differential privacy: Simplifications, extensions, and lower bounds. In: Theory of Cryptography Conference (TCC), pp 635–658, https://doi.org/10.1007/978-3-662-53641-4_24

[5] Fatehi K, Rahmani Ghourtani M, Sonee A, et al (2026) Interpretable attention-based multi-agent PPO for latency spike resolution in 6G RAN slicing. In: Proc. IEEE Int. Conf. Commun. (ICC), https://doi.org/10.1109/ICC59461.2026.11586913

[6] Gong Y, Gong P, Jiang H, et al (2026) Intelligent resource allocation for multiaccess edge computing in 5G ultra-dense slicing network using federated multiagent DDPG. China Commun 23(1):273–289. https://doi.org/10.23919/JCC.fa.2022-0187.202601

[7] Kunasegran MP, Pang WL, Phang SK (2025) Optimising resource allocation in 5G networks: Balancing URLLC and eMBB traffic under gNB congestion. In: 2025 Multimedia University Engineering Conference (MECON), pp 1–6, https://doi.org/10.1109/MECON67253.2025.11277096

[8] Kunasegran MP, Pang WL, Phang SK (2025) SLA-aware DRL for joint slice admission control and load balancing in multi-service 5G RAN. In: 2025 9th International Conference on Recent Advances and Innovations in Engineering (ICRAIE), pp 207–212, https://doi.org/10.1109/ICRAIE65839.2025.11239148

[9] Kunasegran MP, Pang WL, Phang SK (2026) Comprehensive review of optimisation techniques for user-centric distributed network slicing in 5G networks. IEEE Access 14:114630–114648. https://doi.org/10.1109/ACCESS.2026.3716911

[10] Li T, Sahu AK, Zaheer M, et al (2020) Federated optimization in heterogeneous networks. In: Proceedings of Machine Learning and Systems (MLSys), pp 429–450

[11] McMahan B, Moore E, Ramage D, et al (2017) Communication-efficient learning of deep networks from decentralized data. In: Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), pp 1273–1282

[12] Mohajer A, Mirzaei A, Darabi M, et al (2026) Joint SLA-aware task offloading and adaptive service orchestration with graph-attentive multi-agent reinforcement learning. IEEE Trans Netw Service Manag 23:3737–3754. https://doi.org/10.1109/TNSM.2026.3673188

[13] Qasim M, Imran M, Akram F, et al (2025) Transformer and graph neural network-based federated learning for cellular traffic prediction with sustainability analysis. IEEE Access 13:159649–159666. https://doi.org/10.1109/ACCESS.2025.3605380

[14] Shao Y, Li R, Hu B, et al (2021) Graph attention network-based multi-agent reinforcement learning for slicing resource management in dense cellular network. IEEE Trans Veh Technol 70(10):10792–10803. https://doi.org/10.1109/TVT.2021.3103416

[15] Sulaiman M, Moayyedi A, Ahmadi M, et al (2023) Coordinated slicing and admission control using multi-agent deep reinforcement learning. IEEE Trans Netw Service Manag 20(2):1110–1124. https://doi.org/10.1109/TNSM.2022.3222589

[16] Tashman DH, Cherkaoui S (2026) Adversarial attacks in AI-driven RAN slicing: SLA violations and recovery. In: Proc. Int. Wireless Commun. and Mobile Computing (IWCMC), https://doi.org/10.1109/IWCMC69287.2026.11580033

[17] Veličković P, Cucurull G, Casanova A, et al (2018) Graph attention networks. In: International Conference on Learning Representations (ICLR)

[18] Yasin TV, Yu CM, Wang LC (2025) Differential privacy federated edge learning-assisted for securing RAN intelligent controller in O-RAN 6G communications. In: Proc. IEEE VTS Asia Pacific Wireless Commun. Symp. (APWCS), https://doi.org/10.1109/APWCS67981.2025.11151868

[19] Zhang H, Zhou H, Erol-Kantarci M (2022) Federated deep reinforcement learning for resource allocation in O-RAN slicing. In: Proc. IEEE Global Commun. Conf. (GLOBECOM), pp 958–963, https://doi.org/10.1109/GLOBECOM48099.2022.10001658